

DAY 16 - MYSQL 8 TRIGGERS

1. What is a Trigger? A Trigger is a database object that automatically executes when a specific event (INSERT, UPDATE, DELETE) occurs on a table.
-

1. Types of Triggers

A) BEFORE Trigger

- Executes before the event occurs.
- Used for validation or modifying data before saving.

B) AFTER Trigger

- Executes after the event occurs.
 - Used for logging and auditing purposes.
-

1. Trigger Events

2. INSERT Trigger runs when a new record is inserted.
 3. UPDATE Trigger runs when an existing record is updated.
 4. DELETE Trigger runs when a record is deleted.
-

1. Employee Table

```
CREATE TABLE employees ( emp_id INT PRIMARY KEY, emp_name VARCHAR(100), salary DECIMAL(10,2) );
```

1. Audit Table

```
CREATE TABLE employee_audit ( audit_id INT AUTO_INCREMENT PRIMARY KEY, emp_id INT, emp_name VARCHAR(100), deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

1. Create Audit-Log Trigger on DELETE

```
DELIMITER //
```

```
CREATE TRIGGER trg_employee_delete AFTER DELETE ON employees FOR EACH ROW BEGIN INSERT INTO employee_audit(emp_id, emp_name) VALUES (OLD.emp_id, OLD.emp_name); END//
```

DELIMITER ;

1. How It Works
 2. AFTER DELETE trigger runs after a row is deleted.
 3. OLD keyword stores the deleted row values.
 4. Deleted data is saved into employee_audit table.
-

1. Test the Trigger

```
INSERT INTO employees VALUES (1,'Tarun',50000), (2,'Rahul',60000);
```

```
DELETE FROM employees WHERE emp_id = 1;
```

1. Check Audit Table

```
SELECT * FROM employee_audit;
```

Expected Output:

audit_id	emp_id	emp_name	deleted_at
----------	--------	----------	------------

1	1	Tarun	2026-06-10
---	---	-------	------------

1. BEFORE UPDATE Trigger Example

```
DELIMITER //
```

```
CREATE TRIGGER trg_before_update_salary BEFORE UPDATE ON employees FOR EACH ROW BEGIN IF  
NEW.salary < 0 THEN SET NEW.salary = 0; END IF; END//
```

```
DELIMITER ;
```

SUMMARY

- BEFORE trigger executes before an event.
- AFTER trigger executes after an event.
- INSERT, UPDATE, DELETE can activate triggers.
- Audit tables track changes in data.
- DELETE audit triggers help maintain history and records.